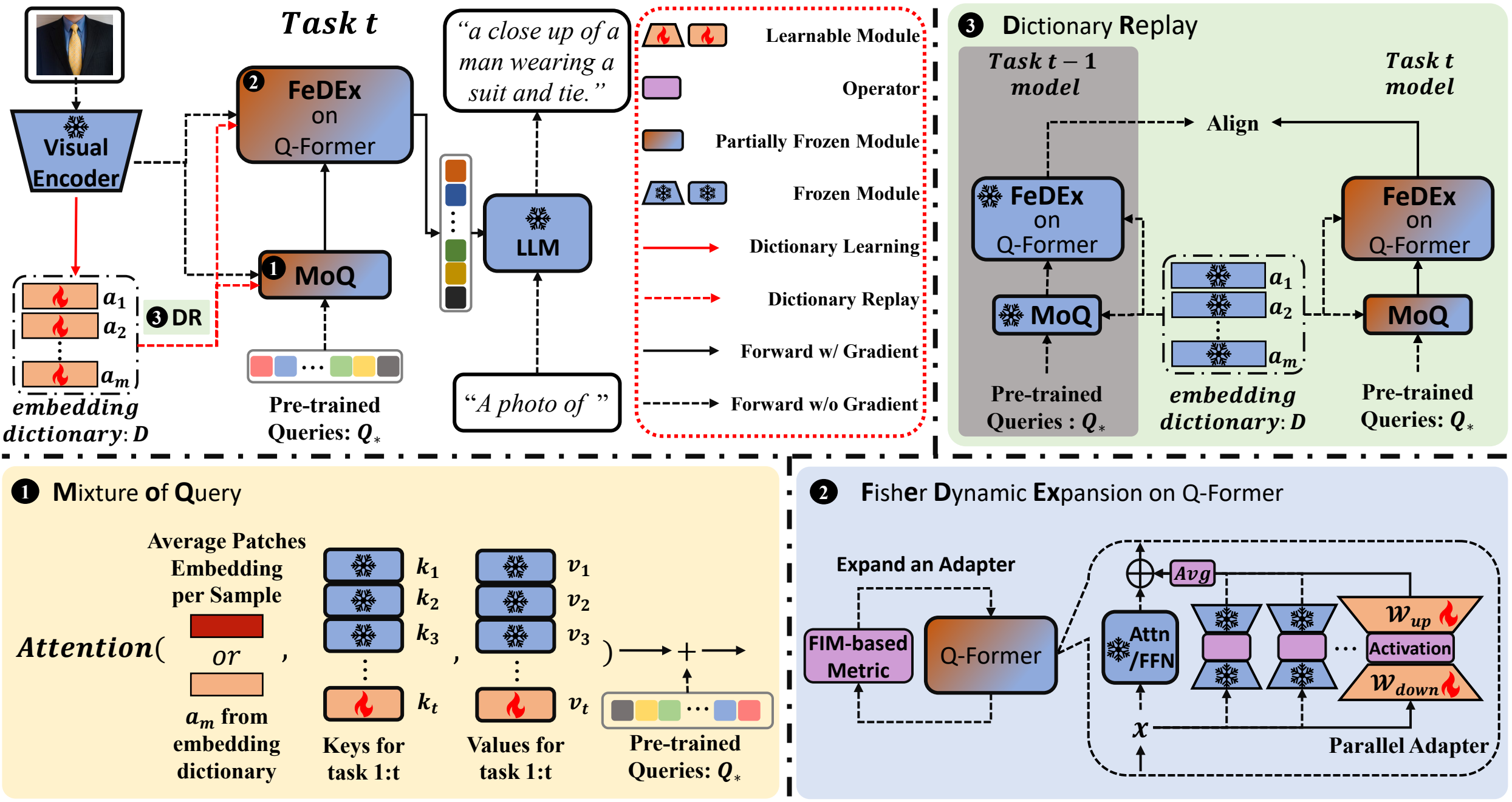
Task t
Visual Encoder
2 FeDEx on Q-Former
1 MoQ
3 DR
LLM
"a close up of a man wearing a suit and tie."
"A photo of "
a_1
a_2
a_m
embedding dictionary: D
Pre-trained Queries: Q_*
Learnable Module
Operator
Partially Frozen Module
Frozen Module
Dictionary Learning
Dictionary Replay
Forward w/ Gradient
Forward w/o Gradient
3 Dictionary Replay
Task t − 1 model
Task t model
Align
FeDEx on Q-Former
MoQ
Pre-trained Queries : Q_*
embedding dictionary: D
Pre-trained Queries: Q_*
1 Mixture of Query
Average Patches Embedding per Sample
Attention(
or
,
a_m from embedding dictionary
k_1
k_2
k_3
k_t
Keys for task 1:t
v_1
v_2
v_3
v_t
Values for task 1:t
)
+
Pre-trained Queries: Q_*
2 Fisher Dynamic Expansion on Q-Former
Expand an Adapter
FIM-based Metric
Q-Former
Attn /FFN
Avg
w_{up}
Activation
w_{down}
x
Parallel Adapter